

Detection and Risk Assessment of Unsafe Hanging Travel in Public Transport Using Machine Learning

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Abstract—Unsafe door-hanging and overcrowded travel are persistent safety risks in Dhaka’s public transport system, particularly in buses and legunas operating under high passenger demand. Existing intelligent transportation research has addressed vehicle detection, passenger counting, and general crowd analysis, but localized automated detection of unsafe hanging travel remains limited. This paper presents an image-based machine-learning framework for classifying public transport scenes into safe/normal and unsafe/risky conditions. A custom bus and leguna image dataset is prepared through cleaning, annotation, preprocessing, and augmentation. For classical machine-learning baselines, Histogram of Oriented Gradients (HOG) descriptors are used with Logistic Regression, Gaussian Naive Bayes, and a Support Vector Machine (SVM) with a radial basis function kernel. Deep models including a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, ConvNeXt-Tiny, and an ensemble are also evaluated to compare learned visual features with handcrafted HOG representations. The retrained dataset contains 432 annotation-labeled images, and augmentation is restricted to the training partition. On the untouched 65-image test set, the ResNet50–ConvNeXt ensemble achieves 86.2% accuracy, 90.5% unsafe recall, and 0.949 ROC-AUC at the balanced operating point.

Index Terms—Unsafe public transport, computer vision, image classification, Histogram of Oriented Gradients, support vector machine, convolutional neural network, intelligent transportation system.

I. INTRODUCTION

Public transport in Dhaka carries a large share of daily urban mobility, but high passenger demand, limited vehicle capacity, and inconsistent enforcement often create unsafe conditions. A common visible hazard is door-hanging travel, where passengers ride from the doorway or exterior boundary of buses and legunas, increasing exposure to falls, collisions, and roadside obstacles.

Safety monitoring still depends heavily on traffic personnel, transport operators, or surveillance staff. Manual observation is difficult to scale across dense road networks, heterogeneous vehicle types, and continuously changing traffic scenes.

Computer vision is useful for this task because images can capture vehicle structure, passenger position, crowding, and doorway conditions. This paper proposes an image-based machine-learning system that classifies Dhaka public transport scenes into safe/normal negative samples and unsafe/risky positive samples.

The main contributions are as follows:

- A localized safe/unsafe public transport image dataset containing Dhaka bus and leguna scenes.
- A complete machine-learning pipeline for unsafe hanging-travel detection.
- A comparison of HOG-based classical machine-learning models and deep convolutional models.
- An evaluation using standard classification metrics, including accuracy, unsafe-class recall, confusion matrices, AUC, and learning curves where applicable.

II. RELATED WORK

Computer vision has become an important component of intelligent transportation systems. Surveyed applications include vehicle detection, pedestrian detection, anomaly detection, traffic monitoring, and automatic number-plate recognition [1]. These methods show that visual data can support automated interpretation of transportation scenes, although most work targets general traffic understanding rather than specific passenger-safety risks.

Passenger monitoring research commonly focuses on counting, occupancy estimation, boarding analysis, and crowd monitoring in buses or terminals [2]. Public transport safety studies identify overcrowding, weak enforcement, and unsafe passenger behavior as important risk factors in dense urban environments [3], while broader reviews show that crowd pressure and non-collision incidents remain persistent concerns when demand exceeds service capacity [4].

Unsafe behavior and abnormal-activity detection are related to the present task because they aim to identify visual patterns that deviate from normal scenes. Recent surveys report that deep learning can be effective for abnormal behavior detection, but they also emphasize difficulties caused by limited labeled data, occlusion, scene variation, and domain transfer [5]. Object-detection methods such as YOLO provide another foundation for transportation surveillance and boarding or alighting recognition [6], [7].

Automatic License Plate Recognition (ALPR) is relevant only as future context for identifying vehicles after unsafe behavior is detected. Existing ALPR studies address plate localization and recognition under real-world imaging constraints [8], [9], [10], including Bengali license-plate recognition under low-resolution and blurred conditions [11].

The key gap is that existing work rarely addresses unsafe door-hanging detection in Dhaka-like crowded public trans-

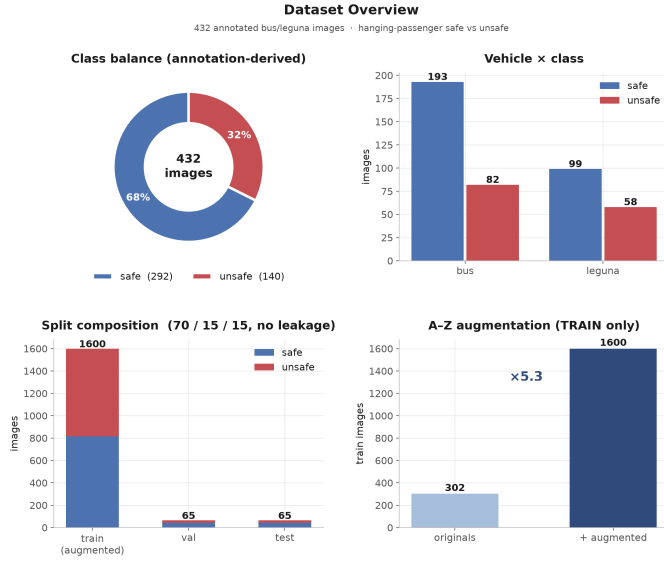


Fig. 1. Updated dataset overview showing vehicle-wise class distribution and stratified 70/15/15 split composition.

TABLE I
DATASET COMPOSITION AFTER SPLITTING AND TRAIN-ONLY AUGMENTATION

Partition	Total	Safe	Unsafe
Training (augmented)	1,600	816	784
Validation	65	44	21
Test	65	44	21
Labeled originals	432	292	140

port environments using localized datasets under real-world challenges such as occlusion, motion blur, lighting variation, crowding, and non-standard vehicle structures. Prior systems often estimate occupancy, count passengers, or recognize boarding events, but they do not directly model hanging passengers as a safety-critical positive class under local bus and leguna conditions.

III. DATASET

A. Data Collection

The dataset contains images of buses and legunas captured under real-world public transport conditions in Dhaka. The samples cover safe/normal and unsafe/risky cases with variation in vehicle type, passenger density, camera angle, lighting, and scene clutter. Labels are derived from image annotations: an image is unsafe if it contains at least one box labeled *unsafe*; otherwise it is safe, and license-plate boxes are ignored. The retrained dataset contains 432 labeled originals, with 292 safe samples and 140 unsafe samples. Figure 1 summarizes the updated class distribution and split composition.

Representative examples of the vehicle categories are shown in Fig. 2 and Fig. 3.



Fig. 2. Representative leguna samples.



Fig. 3. Representative bus samples.

B. Data Cleaning

Cleaning removed images whose content was blurred, ambiguous, low quality, heavily obstructed, poorly framed, or unreliable for class assignment. This step reduced label noise while preserving realistic variation in traffic, vehicle structure, crowding, and illumination.

C. Data Annotation

Manual annotation assigned each retained image to one of two classes: negative for safe/normal passenger conditions and positive for unsafe/risky hanging travel. Figure 4 shows representative annotation examples, including license-plate annotations retained only for possible future vehicle-identification integration.

D. Data Augmentation

Augmentation was restricted to the training partition to prevent leakage into validation or test data. The 302 training originals produced 1,298 offline augmented samples, yielding 1,600 training images with 816 safe and 784 unsafe samples. Validation and test each contain 65 unaugmented originals,



(a) Safe/normal annotation.



(b) Unsafe/risky annotation.



(c) License-plate annotation.



(d) Combined annotation.

Fig. 4. Representative annotation examples.

divided into 44 safe and 21 unsafe images. Deep models additionally used online augmentation, weighted sampling, and horizontal-flip test-time augmentation. These transformations simulate common roadside imaging variation while preserving the class label [12].

All images were standardized before model-specific preprocessing. The standardized preprocessing and model input settings are summarized in Table II. The dataset is available online at <https://drive.google.com/drive/folders/1D3jDBvA-cAhwRPA4wUBUss5zIwdXdLJY>.

E. Ethical Considerations

The dataset is used only for academic safety-monitoring research. Images may contain privacy-sensitive visual information such as human faces, vehicle license plates, and location-related context. The classification task focuses on transport safety conditions rather than personal identity recognition; no face recognition, identity matching, or individual profiling is performed. Before any public release or operational deployment, privacy-preserving steps such as face and license-plate anonymization, removal of GPS or location metadata, consent and data-governance procedures, and compliance with relevant regulations would be required.

IV. METHODOLOGY

The proposed method is a binary image-classification pipeline for public transport safety assessment. The workflow consists of dataset preparation, preprocessing, feature extraction or model-specific input preparation, model training, and evaluation. HOG-based classical models receive handcrafted gradient features, while deep neural models learn directly from image tensors. Implementation settings are summarized in Table II.

A. Pipeline Overview

The pipeline begins with Dhaka bus and leguna image collection, followed by cleaning, annotation, augmentation, and standardization. Classical machine-learning models use HOG feature vectors extracted from grayscale inputs, whereas CNN and transfer-learning models receive RGB tensors. All models are evaluated on the same held-out split using standard classification metrics.

B. Training and Inference Procedure

The complete retraining procedure is summarized below so that data preparation, model fitting, and deployment decisions are defined within the methodology.

- 1) **Annotation-based labeling:** Begin with 432 labeled originals containing 292 safe and 140 unsafe images. Assign an image to the unsafe class when at least one annotation box is labeled `unsafe`; ignore license-plate boxes.
- 2) **Four-way stratification:** Split the originals 70/15/15 over vehicle type and class, preserving bus-safe, bus-unsafe, leguna-safe, and leguna-unsafe samples in validation and test partitions.
- 3) **Leakage-free augmentation:** Apply offline augmentation only to the 302 training originals, producing 1,600 training images. Keep the 65-image validation and 65-image test partitions unaugmented.
- 4) **Model-specific preparation:** Extract standardized HOG features followed by PCA for classical models. Supply normalized RGB tensors to the scratch CNN and ImageNet-pretrained transfer-learning models.
- 5) **Model training:** Train Logistic Regression, Gaussian Naive Bayes, SVM, and the scratch CNN as baselines. Fine-tune ResNet18, EfficientNet-B0, ResNet50, and ConvNeXt-Tiny in two phases. Use weighted sampling and class-weighted loss for deep models.
- 6) **Validation and ensembling:** Tune balanced, high-recall, and max-recall decision thresholds on validation predictions. Form the final ensemble by averaging ResNet50 and ConvNeXt-Tiny unsafe-class probabilities.
- 7) **Test-time inference:** Apply horizontal-flip test-time augmentation to the deep models, average the resulting probabilities, select the configured operating threshold, and evaluate once on the untouched test set.

Model 2: SVM with HOG features

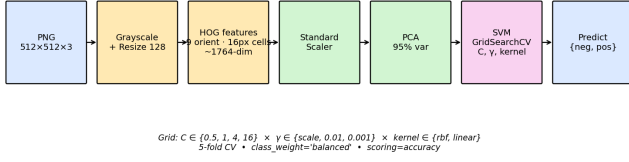


Fig. 5. Overview of the HOG-based classical machine-learning pipeline.

C. HOG-Based Feature Representation

For the classical models, Histogram of Oriented Gradients (HOG) descriptors were extracted from grayscale images [13]. HOG captures local gradient orientation and edge-structure information, which is useful for posture-, shape-, and boundary-related visual classification. Feature extraction was implemented with scikit-image [14]; Table II lists the extraction settings.

D. Classical Machine Learning Models

The classical baselines use HOG features with three complementary classifiers. Logistic Regression provides a linear probabilistic baseline and estimates the posterior probability of the positive class as

$$P(y = 1 | x) = \frac{1}{1 + e^{-w^T x}}.$$

The model is computationally efficient and interpretable, but its linear decision boundary can be limited when safe and unsafe samples are not linearly separable.

Gaussian Naive Bayes is a probabilistic baseline that assumes conditional independence among features given the class label. Each feature is modeled with a Gaussian likelihood,

$$P(x_i | y) = \mathcal{N}(\mu_y, \sigma_y^2),$$

and prediction is based on the posterior score

$$P(y | x) \propto P(y) \prod_i P(x_i | y).$$

Although the independence assumption is restrictive for correlated HOG features, the model is useful as a low-complexity comparator.

The Support Vector Machine (SVM) seeks a maximum-margin boundary between the negative and positive classes [15]. The radial basis function (RBF) kernel handles non-linear separation in the HOG feature space through

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

This kernel allows samples with similar HOG patterns to influence the decision function even when a linear boundary is insufficient. Table II summarizes the implementation settings for all three classical models.

Model 1: Small CNN (from scratch) — ~1.2M params

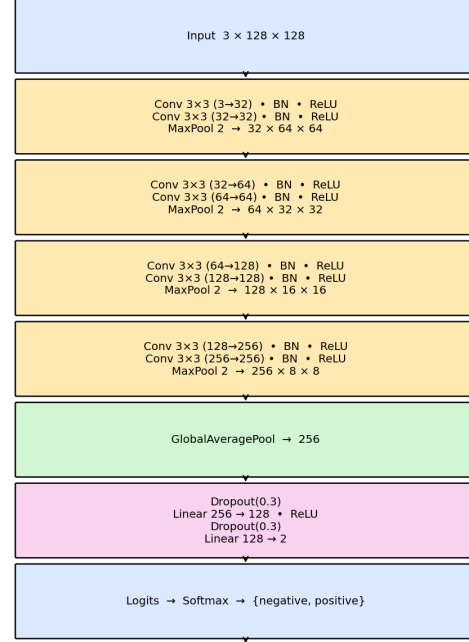


Fig. 6. CNN architecture used for image-level binary classification.

E. Deep Neural Models

The deep models process RGB images directly and learn spatial features from the training data [16]. The experiments evaluate five neural architectures: a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, and ConvNeXt-Tiny. The scratch CNN provides a lightweight learned-feature baseline, while the other architectures use ImageNet-pretrained weights and two-phase fine-tuning [17], [18], [19]. A probability-level ensemble combines ResNet50 and ConvNeXt-Tiny. Weighted sampling, class-weighted loss, validation-tuned thresholds, and horizontal-flip test-time augmentation address imbalance and improve unsafe-class recall. Figure 6 shows the CNN architecture, and Table II reports the combined configuration.

V. EXPERIMENTAL SETUP

A. Split Protocol

The data split followed a four-way stratified protocol over vehicle type and class, producing 70/15/15 training, validation, and test partitions. This ensures that bus-safe, bus-unsafe, leguna-safe, and leguna-unsafe samples occur in both validation and test sets. Augmentation was applied only after splitting and only to training originals. Thresholds were tuned on validation predictions, while the 65-image test set remained untouched until final evaluation.

B. Model Configuration

The experiments were implemented in Python using scikit-learn for the classical pipelines [22], PyTorch for the neural model [23], and scikit-image for image processing [14]. A fixed random seed of 42 was used for reproducibility. Model and preprocessing settings are consolidated in Table II.

For HOG-based models, preprocessing and dimensionality-reduction steps were fitted on the training data and then applied to validation and test data. Deep models used normalized RGB image tensors rather than HOG descriptors.

For neural-model inference, logits $z \in \mathbb{R}^2$ are converted to class probabilities by

$$p_k = \frac{e^{z_k}}{\sum_{j=0}^1 e^{z_j}}, \quad k \in \{0, 1\}.$$

The optimizer and learning-rate schedule are listed in Table II [24].

C. Evaluation Metrics

Evaluation used accuracy, precision, recall, F1-score, confusion matrices, and AUC. Learning curves were used where applicable to inspect neural-model training behavior. Let TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, with the positive class corresponding to unsafe/risky hanging travel. The scalar metrics are

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN}, \\ \text{Precision} &= \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \\ F_1 &= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned}$$

Because missed unsafe cases have direct safety implications, positive-class recall is considered alongside overall accuracy and AUC.

VI. RESULTS

A. Quantitative Comparison

Table III reports balanced operating-point performance on the untouched 65-image test set. The ensemble gives the strongest overall result with 0.862 accuracy, 0.905 unsafe recall, and 0.949 ROC-AUC. Among individual models, ResNet50 and SVM both reach 0.846 test accuracy, while ConvNeXt-Tiny reaches 0.944 AUC and 0.857 unsafe recall.

B. Confusion Matrix Analysis

The confusion matrices in Fig. 10 report class-level outcomes for all nine models on the held-out test set. The rows show true labels and the columns show predicted labels, with safe representing normal passenger conditions and unsafe representing door-hanging risk.

C. ROC and AUC Analysis

AUC values are reported in Table III. Figure 11b shows held-out ROC curves for all reported models, while Fig. 11c shows the corresponding precision-recall curves.

D. Error Analysis

False negatives are the most safety-critical error type because they correspond to unsafe/risky hanging-travel cases predicted as safe/normal. At the balanced operating point, the ensemble detects 90.5% of unsafe test images. The high-recall threshold raises unsafe recall to 100% with 69.2% accuracy, while the max-recall threshold also retains 100% recall but lowers accuracy to 41.5%. These modes expose the expected safety tradeoff between missed unsafe cases and false alarms.

TABLE II
MODEL CONFIGURATION

Model / Stage	Input Representation	Key Configuration
Image standardization	Source images	Orientation correction, RGB conversion, shorter side 512 px, center crop to 512×512 , PNG output.
HOG feature extraction	Grayscale 128×128	9 orientation bins, 16×16 pixel cells, 2×2 cell blocks, L2-Hys normalization, 1764-d feature vector.
Logistic Regression	HOG features	StandardScaler and PCA fitted on training data, followed by Logistic Regression with <code>lbfgs</code> .
Gaussian Naive Bayes	HOG features	StandardScaler and PCA fitted on training data, followed by Gaussian Naive Bayes.
SVM with RBF kernel	HOG features	StandardScaler and PCA fitted on training data, followed by an RBF-kernel SVM with probability estimates enabled.
CNN	RGB 128×128	4-block CNN trained from scratch; updated test metrics are included in the model comparison.
ResNet18	RGB images	ImageNet-pretrained residual network with two-phase fine-tuning, weighted sampling, class-weighted loss, and validation-tuned thresholds.
EfficientNet-B0	RGB images	ImageNet-pretrained compound-scaled network with the same imbalance handling and validation protocol.
ResNet50	RGB images	Deeper ImageNet-pretrained residual network with two-phase fine-tuning and horizontal-flip test-time augmentation.
ConvNeXt-Tiny	RGB images	ImageNet-pretrained modern convolutional network with two-phase fine-tuning and horizontal-flip test-time augmentation.
Ensemble	RGB images	Probability-level ensemble of ResNet50 and ConvNeXt-Tiny with balanced, high-recall, and max-recall validation-tuned thresholds.
Evaluation protocol	Train/validation/test	Four-way stratified 70/15/15 split; 302 training originals expanded to 1,600 train images, with 65 unaugmented validation and 65 untouched test images.

TABLE III
HELD-OUT TEST PERFORMANCE FOR ALL REPORTED MODELS

Model	Acc.	Bal. Acc.	Unsafe R	Unsafe P	Spec.	F1	AUC	PR-AUC	MCC
Ensemble	0.862	0.873	0.905	0.731	0.841	0.809	0.949	0.919	0.712
ResNet50	0.846	0.812	0.714	0.789	0.909	0.750	0.943	0.914	0.641
ConvNeXt-Tiny	0.815	0.826	0.857	0.667	0.795	0.750	0.944	0.915	0.619
EfficientNet-B0	0.769	0.792	0.857	0.600	0.727	0.706	0.909	0.887	0.548
ResNet18	0.831	0.813	0.762	0.727	0.864	0.744	0.892	0.877	0.618
CNN	0.815	0.777	0.667	0.737	0.886	0.700	0.866	0.778	0.569
SVM (RBF)	0.846	0.824	0.762	0.762	0.886	0.762	0.863	0.834	0.648
Logistic Regression	0.785	0.754	0.667	0.667	0.841	0.667	0.800	0.673	0.508
Naive Bayes	0.677	0.612	0.429	0.500	0.795	0.462	0.646	0.576	0.234

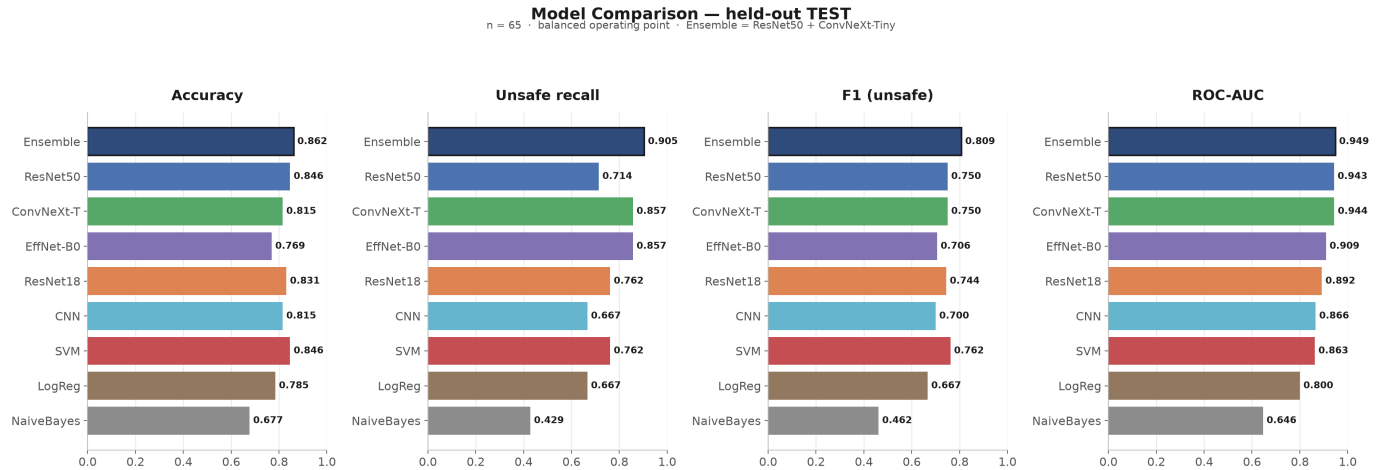


Fig. 7. Held-out test comparison of accuracy, unsafe recall, and ROC-AUC across all nine reported models.

All Metrics at a Glance — TEST

balanced operating point · n = 65

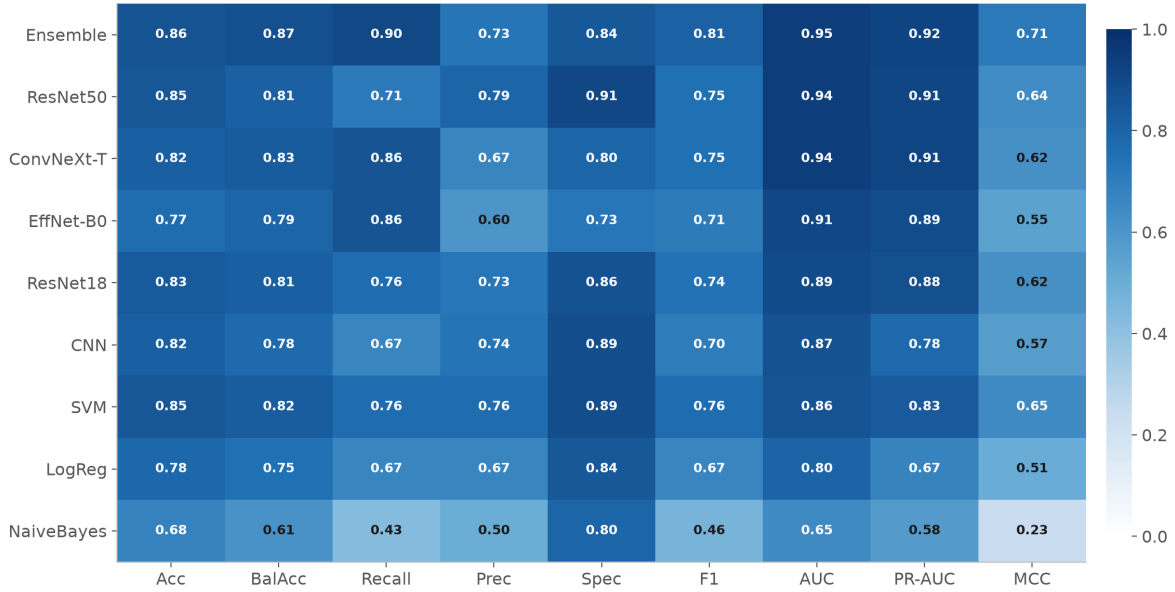


Fig. 8. Heatmap of held-out test metrics for all nine models at the balanced operating point.

Model comparison — held-out test, balanced operating point

Model	Acc	BalAcc	Recall ↑	Prec	Spec	F1	AUC	PR-AUC	MCC
Ensemble	0.862	0.873	0.905	0.731	0.841	0.809	0.949	0.919	0.712
ResNet50	0.846	0.812	0.714	0.789	0.909	0.750	0.943	0.914	0.641
ConvNeXt-T	0.815	0.826	0.857	0.667	0.795	0.750	0.944	0.915	0.619
EffNet-B0	0.769	0.792	0.857	0.600	0.727	0.706	0.909	0.887	0.548
ResNet18	0.831	0.813	0.762	0.727	0.864	0.744	0.892	0.877	0.618
CNN	0.815	0.777	0.667	0.737	0.886	0.700	0.866	0.778	0.569
SVM	0.846	0.824	0.762	0.762	0.886	0.762	0.863	0.834	0.648
LogReg	0.785	0.754	0.667	0.667	0.841	0.667	0.800	0.673	0.508
NaiveBayes	0.677	0.612	0.429	0.500	0.795	0.462	0.646	0.576	0.234

Fig. 9. Visual summary of the complete held-out test results corresponding to Table III.

TABLE IV
SELECTABLE OPERATING POINTS FOR THE DEPLOYED ENSEMBLE.

Mode	Threshold	Test Acc.	Unsafe R	Unsafe P	Spec.
Balanced	0.175	0.862	0.905	0.731	0.841
High recall	0.035	0.692	1.000	0.512	0.545
Max recall	0.013	0.415	1.000	0.356	0.136

Confusion Matrices — TEST

balanced operating point, n = 65

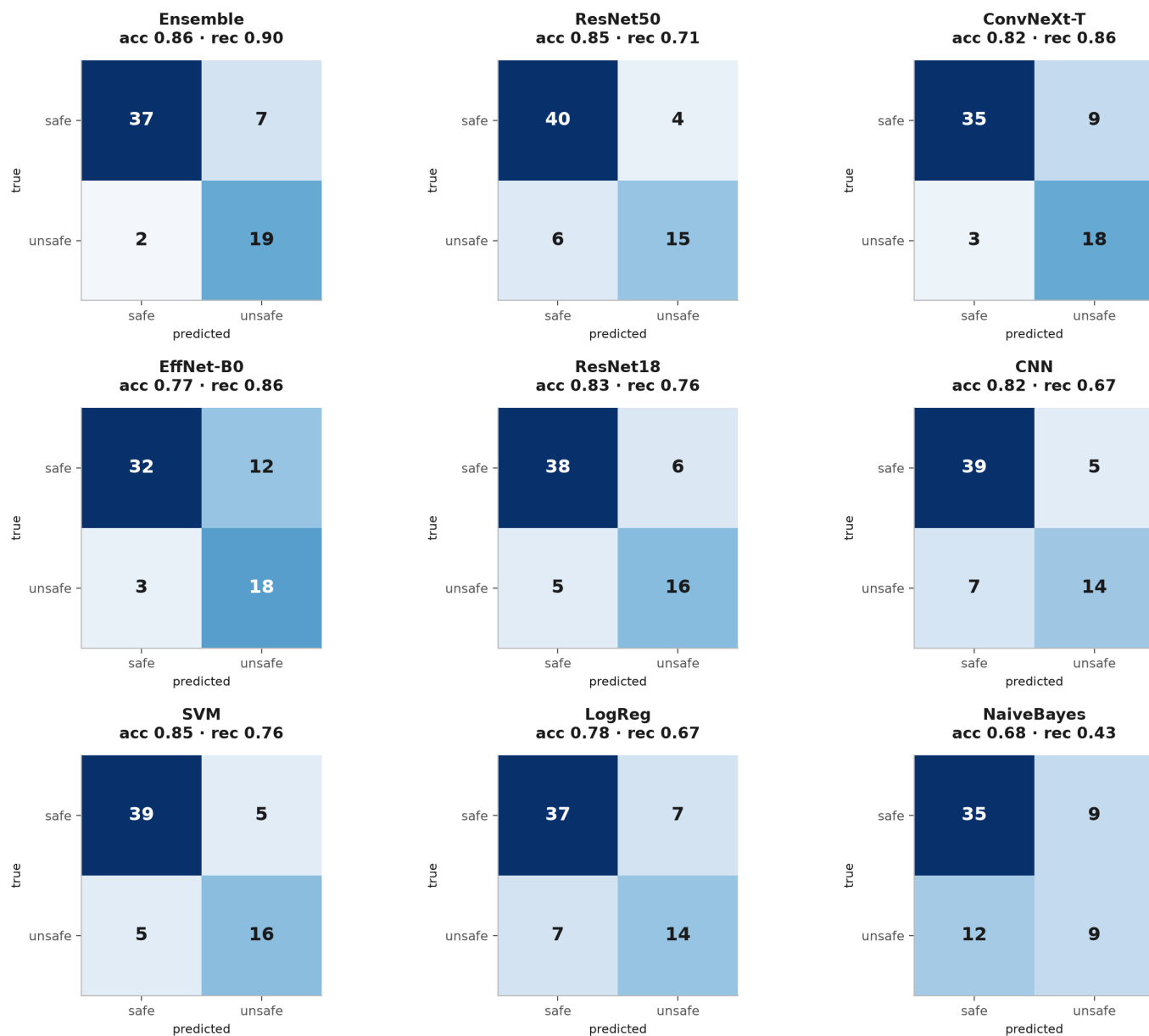
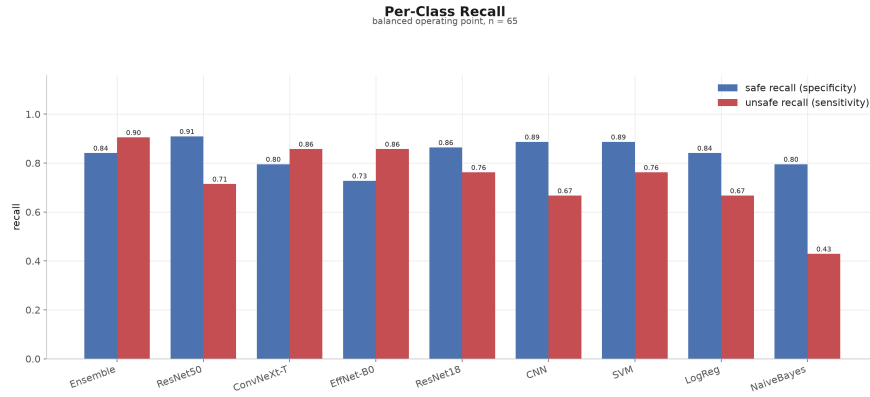
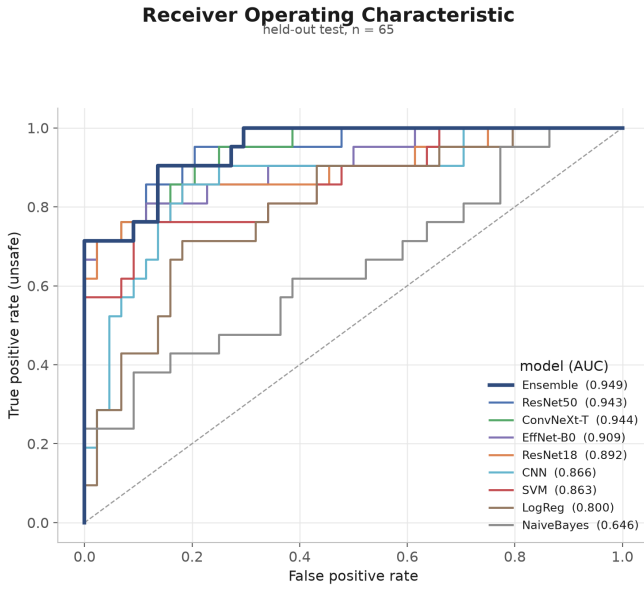


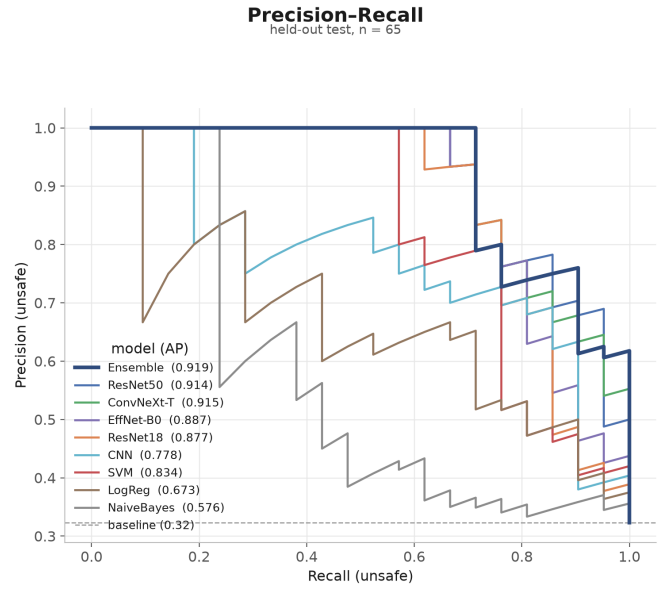
Fig. 10. Confusion-matrix grid for all nine reported models on the held-out test set.



(a) Safe- and unsafe-class recall at the balanced operating point.

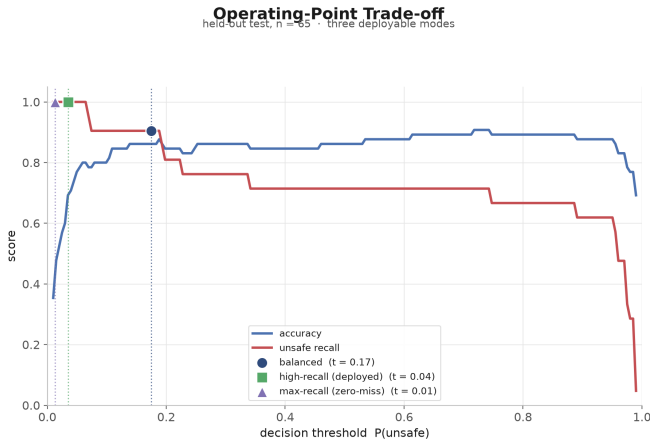


(b) ROC curves.

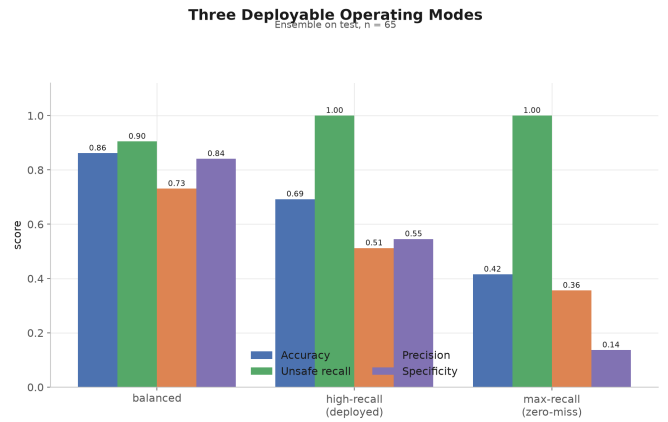


(c) Precision-recall curves.

Fig. 11. Class-wise recall and discrimination curves for all nine models on the held-out test set.



(a) Threshold tradeoff for accuracy, unsafe recall, and precision.



(b) Balanced, high-recall, and max-recall operating modes.

Fig. 12. Ensemble threshold selection and deployable operating modes on the held-out test set.

VII. DISCUSSION

A. Interpretation of Results

The retrained ensemble provides the strongest balanced operating point, reaching 86.2% test accuracy, 90.5% unsafe recall, 0.809 unsafe F1, 0.949 ROC-AUC, and 0.919 PR-AUC. The additional unsafe training data raises AUC beyond the previous 0.929 result and substantially improves balanced unsafe recall.

Among individual deep models, ConvNeXt-Tiny provides 85.7% unsafe recall and 0.944 AUC, while ResNet50 gives 84.6% accuracy and the highest unsafe precision at 78.9%. Among classical models, SVM remains the strongest HOG baseline, reaching 84.6% accuracy, 76.2% unsafe recall, and 0.863 AUC.

Gaussian Naive Bayes underperforms because its conditional-independence assumption is poorly matched to HOG descriptors. Neighboring gradient bins and block-normalized features are correlated by construction, so treating them as independent weakens the model's ability to distinguish visually similar safe/normal and unsafe/risky scenes.

The train-to-test gaps in Table V range from 11.3 to 20.7 percentage points. SVM has the smallest gap, while EfficientNet-B0 has the largest. These gaps indicate that the expanded but still modest set of unique originals does not eliminate overfitting, even though the ensemble improves final discrimination.

For this application, false negatives are more consequential than ordinary classification errors because they correspond to missed risky travel conditions. Precision and overall accuracy remain relevant for reducing false alarms, but a safety-oriented monitoring system should prioritize reducing missed unsafe cases while maintaining acceptable discrimination across normal and risky scenes.

B. Limitations

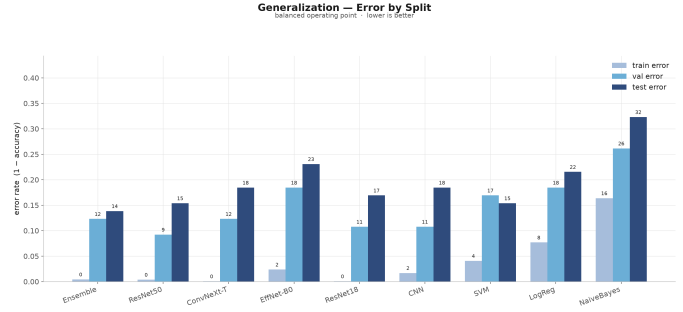
The main limitation of this study is the limited number of unique source images. The updated dataset contains 432 labeled originals with 140 unsafe cases, which is still not enough to fully cover the diversity of vehicle layouts, camera viewpoints, passenger postures, weather, illumination, and traffic environments encountered in deployment.

The neural models also depend on training-time augmentation to increase effective sample diversity. Augmentation improves robustness to controlled transformations, but it does not replace additional real images containing natural occlusion, motion blur, dense crowding, and non-standard bus or leguna structures. These factors can make unsafe/risky positive cases difficult to distinguish from visually cluttered safe/normal scenes.

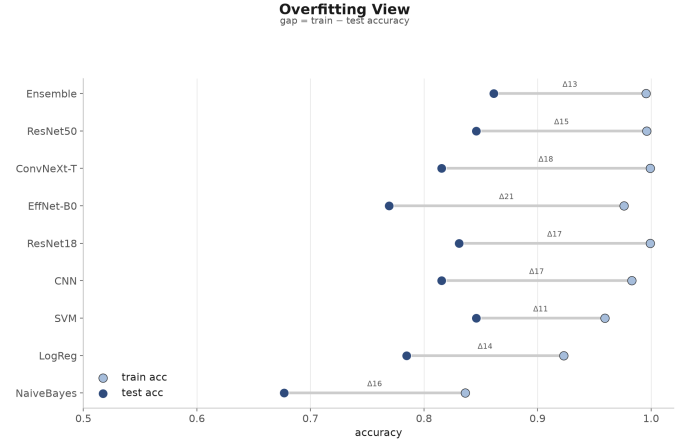
The current system is limited to image-level binary classification. It does not localize hanging passengers, estimate crowd density, track passengers across frames, or evaluate performance on large-scale real-time video streams. Roadside transport images may also include identifiable people,

TABLE V
TRAIN, VALIDATION, AND TEST GENERALIZATION

Model	Train Acc.	Train Err.	Val Acc.	Val Err.	Test Acc.	Test Err.	Test R
Ensemble	0.996	0.004	0.877	0.123	0.862	0.138	0.905
ResNet50	0.996	0.004	0.908	0.092	0.846	0.154	0.714
ConvNeXt-Tiny	0.999	0.001	0.877	0.123	0.815	0.185	0.857
EfficientNet-B0	0.976	0.024	0.815	0.185	0.769	0.231	0.857
ResNet18	0.999	0.001	0.892	0.108	0.831	0.169	0.762
CNN	0.983	0.017	0.892	0.108	0.815	0.185	0.667
SVM (RBF)	0.959	0.041	0.831	0.169	0.846	0.154	0.762
Logistic Regression	0.923	0.077	0.815	0.185	0.785	0.215	0.667
Naive Bayes	0.836	0.164	0.738	0.262	0.677	0.323	0.429



(a) Train, validation, and test error.



(b) Train-to-test generalization gap.

Fig. 13. Generalization behavior of all nine models after retraining.

license plates, and location cues, so deployment would require anonymization and data-governance safeguards. The method should therefore be interpreted as a baseline for visual safety classification rather than an operational monitoring system.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a computer-vision and machine-learning pipeline for detecting unsafe hanging travel in Dhaka public transport. The task was formulated as binary image classification over safe/normal and unsafe/risky passenger conditions. A localized bus and leguna dataset was prepared through cleaning, annotation, preprocessing, and augmentation, with HOG features used for classical models and RGB inputs used for CNN-based classification.

The updated result set compares Logistic Regression, Gaussian Naive Bayes, SVM with an RBF kernel, a CNN trained from scratch, ResNet18, EfficientNet-B0, ResNet50, ConvNeXt-Tiny, and a ResNet50–ConvNeXt ensemble. On the untouched test set, the ensemble achieves the strongest balanced result with 86.2% accuracy, 90.5% unsafe recall, and 0.949 ROC-AUC. The deployed high-recall mode reaches 100% unsafe recall at 69.2% accuracy, while max-recall mode also preserves 100% recall at 41.5% accuracy.

As a practical outcome of this work, we developed an online platform that allows users to classify public-transport images as safe/normal or unsafe/risky. The platform is available at `ml-predictor-app.streamlit.app`.

Future work should focus on collecting a larger localized dataset, improving recall for unsafe hanging travel, evaluating the system on real-time video streams, and designing privacy-aware deployment procedures. Detection-based models such as YOLO [6] should also be tested to localize hanging passengers rather than only classify whole images. License-plate recognition may be integrated only as a future extension for vehicle identification after unsafe conditions are detected.

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